

MODULE-2

PROGRAMMING PRACTICE USING JAVA

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SECTION: - A8

YEAR: - 3rd

TOPICS COVERED: -

- .LINEAR SEARCH**
- .BINARY SEARCH**
- .BUBBLE SORT**
- .INSERTION SORT**
- .SELECTION SORT**

MODULE - 2

1. Fraud Transaction Detection (Linear Search)

Code:

```
import java.util.Scanner;
public class Fraud Transaction {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        boolean found = false;
        for (int i = 0; i < n; i++) {
            if (arr[i] == target) {
                found = true;
                break;
            }
        }
        if (found) {
            System.out.println("Fraud Transaction Found");
        } else {
            System.out.println("Transaction Not Found");
        }
        sc.close();
    }
}
```



```

1 package module;
2 import java.util.Scanner;
3 public class FraudTransaction {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int arr[] = new int[n];
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        int target = sc.nextInt();
12        boolean found = false;
13        for (int i = 0; i < n; i++) {
14            if (arr[i] == target) {
15                found = true;
16                break;
17            }
18        }
19        if (found) {
20            System.out.println("Fraud Transaction Found");
21        } else {
22            System.out.println("Transaction Not Found");
23        }
24        sc.close();
25    }
26 }
27

```

@ Javadoc Declaration Console X

<terminated> FraudTransaction [Java Application] C:\Users\mahil\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-06

```

5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found

```

2. Server Log Search (Binary Search)

```
code: import java.util.Scanner;
public class ServerLogSearch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        int low = 0;
        int high = n - 1;
        boolean found = false;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (arr[mid] == target) {
                found = true;
                break;
            } else if (arr[mid] < target) {
                low = mid + 1;
            } else {
                high = mid - 1;
            }
        }
        if (found) {
            System.out.println("Log Found");
        } else {
            System.out.println("Log Not Found");
        }
        sc.close();
    }
}
```



```

1 package module;
2 import java.util.Scanner;
3 public class ServerLogSearch {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int arr[] = new int[n];
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        int target = sc.nextInt();
12        int low = 0;
13        int high = n - 1;
14        boolean found = false;
15        while (low <= high) {
16            int mid = (low + high) / 2;
17            if (arr[mid] == target) {
18                found = true;
19                break;
20            } else if (arr[mid] < target) {
21                low = mid + 1;
22            } else {
23                high = mid - 1;
24            }
25        }
26        if (found) {
27            System.out.println("Log Found");
28        } else {
29            System.out.println("Log Not Found");
30        }
31        sc.close();
32    }
33 }

```

@ Javadoc Declaration Console X

<terminated> ServerLogSearch [Java Application] C:\Users\mahi\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0

```

7
1001 1005 1010 1015 1020 1030 1040
1015

```

Log Found

3. Sort Employee Salaries (Bubble Sort)

Code:

```
import java.util.Scanner;
public class BubblesortSalary {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for(int i = 0; i < n - 1; i++) {
            for(int j = 0; j < n - i - 1; j++) {
                if(arr[j] > arr[j+1]) {
                    int temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                }
            }
        }
        for(int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

```

1 package module;
2 import java.util.Scanner;
3 public class BubbleSortSalary {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int arr[] = new int[n];
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        for (int i = 0; i < n - 1; i++) {
12            for (int j = 0; j < n - i - 1; j++) {
13                if (arr[j] > arr[j + 1]) {
14                    int temp = arr[j];
15                    arr[j] = arr[j + 1];
16                    arr[j + 1] = temp;
17                }
18            }
19        }
20        for (int i = 0; i < n; i++) {
21            System.out.print(arr[i] + " ");
22        }
23        sc.close();
24    }
25 }

```

@ Javadoc Declaration Console X

<terminated> BubbleSortSalary [Java Application] C:\Users\mahi\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000

4. Sort Website Response Time (Insertion Sort)

```
Code:- import java.util.Scanner;
public class InsertionSortResponse {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 1; i < n; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```



```

1 package module;
2 import java.util.Scanner;
3 public class InsertionSortResponse {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int arr[] = new int[n];
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        for (int i = 1; i < n; i++) {
12            int key = arr[i];
13            int j = i - 1;
14            while (j >= 0 && arr[j] > key) {
15                arr[j + 1] = arr[j];
16                j--;
17            }
18            arr[j + 1] = key;
19        }
20        for (int i = 0; i < n; i++) {
21            System.out.print(arr[i] + " ");
22        }
23        sc.close();
24    }
25 }

```

@ Javadoc Declaration Console X

<terminated> InsertionSortResponse [Java Application] C:\Users\mahil\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-

```

5
250 120 320 180 100
100 120 180 250 320

```

5. Prioritize Bug Reports (Selection Sort)

Code :-

```
import java.util.Scanner;
public class SelectionSortBug {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            int min = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[min]) {
                    min = j;
                }
            }
            int temp = arr[i];
            arr[i] = arr[min];
            arr[min] = temp;
        }
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```



```

1 package module;
2 import java.util.Scanner;
3 public class SelectionSortBug {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int arr[] = new int[n];
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        for (int i = 0; i < n - 1; i++) {
12            int min = i;
13            for (int j = i + 1; j < n; j++) {
14                if (arr[j] < arr[min]) {
15                    min = j;
16                }
17            }
18            int temp = arr[i];
19            arr[i] = arr[min];
20            arr[min] = temp;
21        }
22        for (int i = 0; i < n; i++) {
23            System.out.print(arr[i] + " ");
24        }
25        sc.close();
26    }
27 }

```

@ Javadoc Declaration Console X

<terminated> SelectionSortBug [Java Application] C:\Users\mahil\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-06:

5
9 2 8 1 5
1 2 5 8 9